

ASSESSMENT 1 BRIEF	
Subject Code and Title	MLN 601 Machine Learning
Assessment	Regression Analysis
Individual/Group	Individual
Length	1,000 words (+/-10%)
Learning Outcomes	The Subject Learning Outcomes demonstrated by successful completion of the task below include: b) Apply learning algorithms to perform machine learning tasks. c) Implement practical machine learning: data pre-processing, analysis, model selection, and interpret the results. d) Communicate clearly and effectively using the technical language of machine learning to a range of stakeholders
Submission	Due by 11:55 pm AEST/AEDT on the Sunday at the end of Module 4.
Weighting	20%
Total Marks	100 marks

Task Summary

In this Assessment, you will use a linear regression Machine Learning (ML) algorithm to analyse data and draw conclusions. To help you create and document the ML model and the results, you will follow the end-to-end Cross-Industry Standard Process for Data Mining (CRISP-DM) (Chapman et al., 2000) methodology. Further, to guide you through the analysis and the writing of the report, a template for your Jupyter Notebook has been provided.

The CRISP-DM template is a necessary resource for the completion of this Assessment. You must consult the CRISP-DM template for further details and information.

Please refer to the Task Instructions (below) for further details on how to complete this task.

Context

In this Assessment, you will complete an end-to-end ML exercise using real-world data. In your future workplaces, you will often be expected to undertake similar exercises using suitable data sets. The template and the use of a methodology will ensure that you do not simply perform an analysis and present your results; rather, you will be able to share the output of the analysis and discuss why and how you adopted the methodology with your work colleagues.

For this Assessment, the practice data set is available from the [UCI ML repository, which contains](https://archive.ics.uci.edu/ml/index.php) nearly 500 real-world data sets (<https://archive.ics.uci.edu/ml/index.php>). Your focus will be on the



[wine quality data set](https://archive.ics.uci.edu/ml/datasets/wine+quality) (<https://archive.ics.uci.edu/ml/datasets/wine+quality>). This data set provides wine quality data across 11 traits, including acidity, residual sugar and alcohol concentration. Importantly, this Assessment requires you to develop a model to predict wine quality on a score between 1 to 10.

You must consider the CRISP-DM from the outset. As the report template indicates, the first stage is Business Understanding. This stage requires consideration of the project problem. For example, it may be easiest to focus on predicting only the quality of the [red wine](https://archive.ics.uci.edu/ml/machine-learning-databases/wine-quality/) rather than that of both the red and white wine (<https://archive.ics.uci.edu/ml/machine-learning-databases/wine-quality/>). Ensure that you clearly document whatever target you select for the problem at the commencement of your report (as per the template).

Task Instructions

You will use your Jupyter Notebook on the Microsoft Azure ML platform or Google Colab and Python 3.6 as the language for all three assessments.

Ultimately, the Notebook will contain both your ML code, data and report documentation.

Your Assessment will be evaluated based on the major stages of the CRISP-DM process as set out in the Notebook template with prompts. The process comprises:

1. Business Understanding;
2. Data Understanding;
3. Data Preparation;
4. Modelling;
5. Evaluation; and
6. Deployment.

The six multi-step stages of the CRISP-DM must be undertaken to complete this Assessment. Note: For ease of working and to complete this Assessment, you should document what you are doing in your Notebook as you progress through the activities (e.g., the steps undertaken and the rationale for the selection of the code). The template will prompt you on how to work through the end-to-end ML process.

Stage 1: Business Understanding

1. This section serves as an introduction. You should write a clear and concise narrative, expressing what you are trying to achieve with regards to your evaluation criteria. Think in terms of ML; for example, the prediction algorithm, the data set selected, what you are seeking from the data set and how you intend to understand the value of your prediction capability.
2. Assess the current situation. See 1.1 of the CRISP-DM template (1.1).



Stage 2: Data Understanding

1. Acquire the relevant wine quality data set from the UCI repository for your prediction model (<https://archive.ics.uci.edu/ml/datasets/wine+quality>). Explicitly specify the data source by providing a specific link and the name of the data set (e.g., red wine, white wine or both) and the method of acquisition (e.g., direct from the URL or a download of the .csv file). The steps taken need to be clearly stated. (2.1).
2. Read this data set into your Notebook. (2.1).
3. Describe the data set inclusive of variables, units and levels. (2.2).
4. Verify the data quality by analysing the data set for structure and missing data. (2.3).
5. Conduct an initial data exploration using data visualisation, reporting and querying the data. (2.4).
6. Use the pairplot function in seaborn to determine the relationship, if any, between the variables. Include the output or the visualisation of the pairplot function in your Notebook and comment on it. (2.4.2).

Stage 3: Data Preparation

1. Select the data that you will use for the analysis. (3.1).
2. Clean the data you have selected to improve the quality of the data. (3.2).

Stage 4: Modelling

1. For this Assessment, you are required to use the linear regression model.
2. Import the linear regression model into your code. (4.1).
3. Record any modelling assumptions. (4.2).
4. Run your model over the data set. (4.3).
5. Record the parameter settings, your rationale for your choice of values and the actual model generated. (4.3).
6. Revise any parameter settings for subsequent model runs. Document all the revisions until the best model is reached. (4.4).

Stage 5: Evaluation

1. Assess the ML results. Ensure you include a statement as to whether the model meets the evaluation criteria.

Stage 6: Deployment

1. For this Assessment, you are not required to deploy your model. For this stage, simply include any lessons that you learned and that you wish to share in relation to the things that went right and wrong, the areas in which you did well and in which you could improve. You can also detail any of your other experiences in completing this Assessment.



Submission Instructions

You are expected to submit two files: 1) A Notebook file with an extension .ipynb; and 2) The direct PDF download via LaTeX from the Notebook. The PDF file will contain the Markdown generated report and your code and results for this ML exercise. Name the files as follows:

MLN601LastNameFirstNameBrief1.ipynb; and

MLN601LastNameFirstNameBrief1.pdf

IMPORTANT NOTE: Submit a text file (.txt) that includes the code(s) of your programs in your submissions.

Submit the files via the Assessment link in the main navigation menu in **MLN601—Machine Learning**.

The Learning Facilitator will provide feedback via the Grade Centre in the LMS portal. Feedback can be viewed in My Grades.

References

Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C. & Wirth, R. (2000). *CRISP-DM 1.0—Step-by-step data mining guide*. Copenhagen, Denmark: SPSS Inc. <https://www.the-modeling-agency.com/crisp-dm.pdf>

Cortez, P., Cerdeira, A., Almeida, F., Matos, T. & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems, Elsevier*, 47(4), 547–553. <https://archive.ics.uci.edu/ml/data sets/wine+quality>

UCI (2019). *Machine learning repository*. Irvine, CA: University of California, School of Information and Computer Science. <http://archive.ics.uci.edu/ml>

Academic Integrity Declaration

I declare that except where I have referenced, the work I am submitting for this assessment task is my own work. I have read and am aware of Torrens University Australia Academic Integrity Policy and Procedure viewable online at <http://www.torrens.edu.au/policies-and-forms>

I am aware that I need to keep a copy of all submitted material and their drafts, and I will do so accordingly.

Assessment Rubric

Assessment Attributes	Fail (Yet to Achieve Minimum Standard) 0–49%	Pass (Functional) 50–64%	Credit (Proficient) 65–74%	Distinction (Advanced) 75–84%	High Distinction (Exceptional) 85–100%
Business Understanding 10%	Not really attempted, incomplete and difficult to comprehend. Problem statement unclear.	Demonstrates an attempt to frame the problem; for example, ‘This Notebook contains the steps for predicting red wine quality according to the values of 11 traits’.	Contains a clear explanation of the motivation to undertake ML. Further, the problem statement recognises the prediction is an oversimplification.	The entire introduction to the report is clear and the desire to predict red wine quality is clearly stated.	Complete, readable and easy to follow through each stage. Additional reference materials suggested. An attempt has been made to understand the value of putting in place the prediction to the business. Consideration of going beyond just red wine prediction.
Data understanding 20%	Entirely incorrect and not completely attempted. Totally lacking any interpretation or discussion of the independent and dependent variables. Fails to specify the source of the data set or the number of observations.	Partially correct but missing plots and poorly labelled. Provides limited comments on the source of the data and variables. Does not explicitly state the dependent variable. Specifies the original source of the data set.	Plots and analysis partially complete, including the correlation analysis. Makes clear the source of the data and the dependent variable (i.e., the quality of the wine). Minor omissions in interpretation.	The analysis describing the data and plots is complete and includes a visualisation of the relationships between the attributes. Comments reflect an understanding of the effects of the attributes on the quality of the wine.	The data understanding has been explicitly tied to the data analysis and the context. All the plots convey information and include adequate references to the data set. Demonstrates a desire to understand more about wine. Includes citations to any additional readings.

Data preparation 10%	Comments missing and appears to be incomplete.	Partially complete. The correlation matrix visualises the variables; however, there is little or no commentary on the positive or negative correlations.	Covers data quality and discusses the data cleaning steps that can be undertaken. The positive and negative correlations are discussed. The variables that have the greatest effects on prediction are noted.	The discussion of the data preparation is mostly made explicit with minor omissions. The positive and negative correlations are discussed. The variables that have the greatest effect on prediction are noted as are those that have the least effect.	The discussion is clear regarding the correlations and appears to be correct in relation to the train/test data. Some additional commentary suggestions. For example, the missing data can be estimated.
Modelling and Evaluation 20%	Overly simplistic and incomplete attempt at modelling. Little or no explanatory notes accompany the modelling.	An appropriate analysis of the ML code but fails to include model assessment comments.	The modelling generation and assessment are mainly correct but include minor errors or omissions.	The modelling is complete but includes minor errors and lacks some depth in relation to interpretation.	The modelling is complete and includes informative comments and interpretations.
Deployment 20%	No real attempt to draw a conclusion detailing the lessons learned or experiences (no or very little reflection).	The conclusion is relatively well structured but fails to detail the lessons learned or discuss the experiences.	The conclusion details the lessons learned but contains no or very little reflection.	The conclusion details the lessons learned and discusses the experience.	The conclusion clearly details the lessons learned and discusses the experience. The conclusion has some depth (e.g., suggests what could be done better next time). Some additional references on wine quality have been considered.

Deliverables: • Jupyter code • Pdf 20%	Insufficient comments made in the markdown reporting. Jupyter Notebook file or pdf not provided. Did not attempt to use the template provided.	Notebook file requires major changes to run. Markdown is hard to follow and confusing.	Notebook runs with minor changes. Markdown is reasonable and can be followed at each step. Pdf file is similar to or the same as the Notebook.	All files run and the Notebook reproduces the submitted file. The commentary at each stage is minimal but it is sufficiently complete to convey the entire CRISP-DM cycle.	The markdown report documents the entire end-to-end process and makes full use of the CRISP-DM stages as highlighted in the template. The Notebook file runs with no errors and matches the pdf. The output touches all the key stages of the CRISP-DM and concludes with comments on the student's own experiences and suggestions for improvement.

The following Subject Learning Outcomes are addressed in this assessment	
SLO b)	Apply learning algorithms to perform machine learning tasks.
SLO c)	Implement practical machine learning: data pre-processing, analysis, model selection and interpret the results.
SLO d)	Communicate clearly and effectively using the technical language of machine learning to a range of stakeholders